Throughput QOS

2026-03-11-19-07-51

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions.By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels,ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

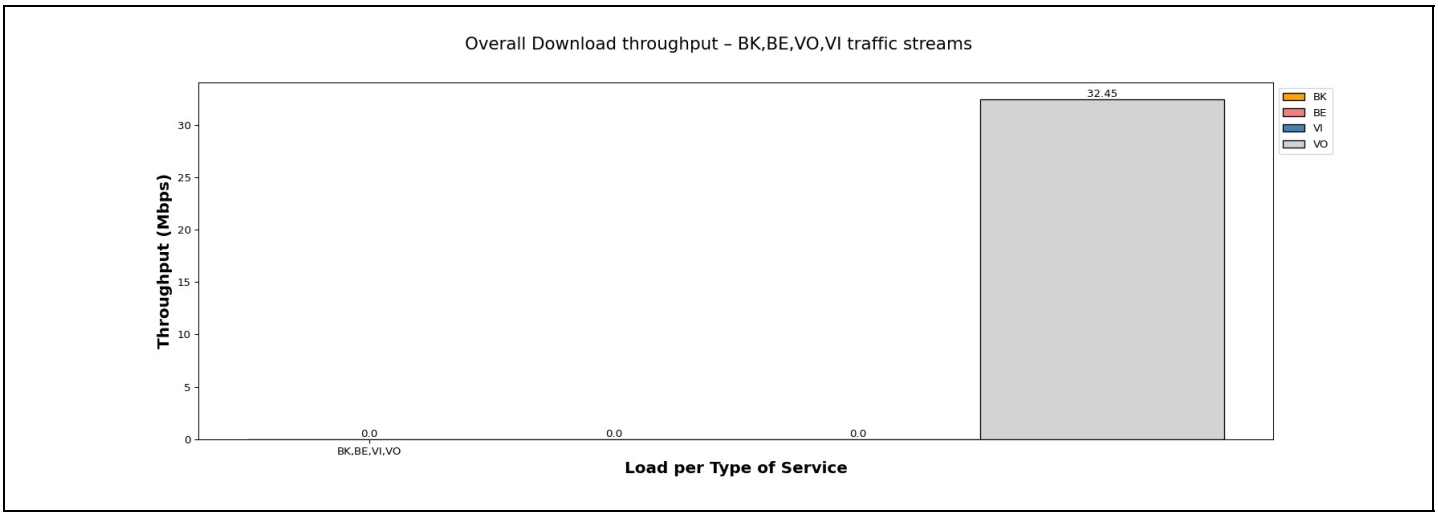
Test Configuration	Number of Stations	32
	AP Model	NETGEAR
	SSID_2.4GHz	NETGEAR_2G_wpa2
	SSID_5GHz	None
	SSID_6GHz	None
	Traffic Duration in hours	0.02
	Security_2.4GHz	wpa2
	Security_5GHz	None
	Security_6GHz	None
	Protocol	UDP
	Traffic Direction	Download
	TOS	['VO']
	Per TOS Load in Mbps	1.0 Mbps

Overall Download throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 1.0 Mbps-download
32	BK : 0.0, BE : 0.0, VI: 0.0, VO: 32.45

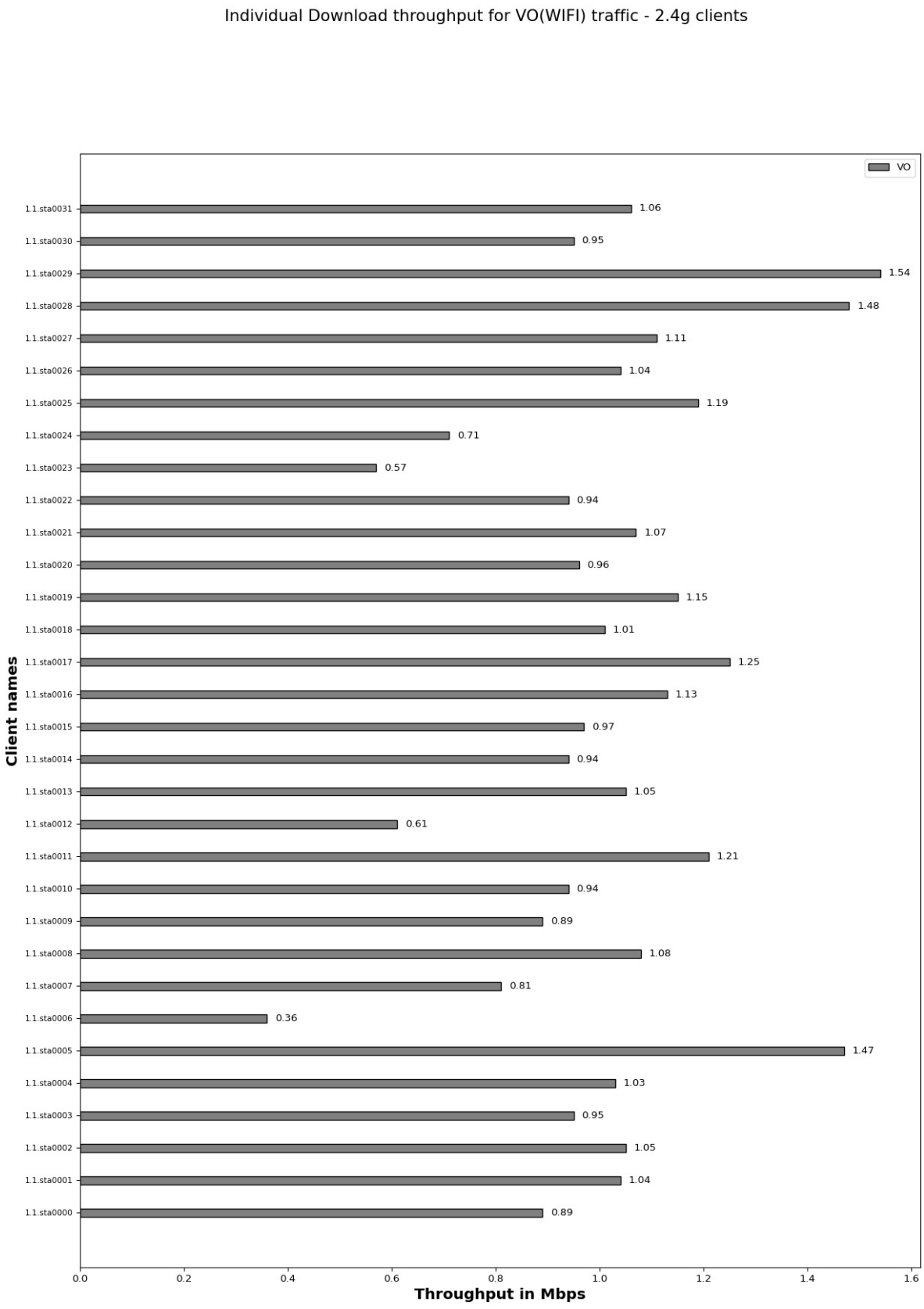
Overall Download throughput for 32 clients for 1.0 Mbps-download band with different TOS.

The below graph represents overall Download throughput for all connected stations running BK, BE, VO, VI traffic with different intended loads per station – 1.0 Mbps-download



Individual Download throughput with intended load 1.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 32 clients running VO (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.



TOS : Voice

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:fb:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.89	0.134567

1.1.sta0001	04:f0:21:94:3e:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.04	0.101574
1.1.sta0002	04:f0:21:94:31:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.05	0.203390
1.1.sta0003	04:f0:21:94:20:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.95	0.000000
1.1.sta0004	04:f0:21:94:19:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.03	38.953093
1.1.sta0005	04:f0:21:94:95:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.47	0.408163
1.1.sta0006	04:f0:21:94:0f:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.36	29.728327
1.1.sta0007	04:f0:21:94:15:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.81	0.000000
1.1.sta0008	04:f0:21:94:24:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.08	0.733038
1.1.sta0009	04:f0:21:94:c4:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.89	0.000000
1.1.sta0010	04:f0:21:94:fc:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.94	37.379706
1.1.sta0011	04:f0:21:94:41:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.21	0.374723
1.1.sta0012	04:f0:21:94:8f:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.61	1.487434
1.1.sta0013	04:f0:21:94:cc:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.05	0.153951
1.1.sta0014	04:f0:21:94:68:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.94	0.000000
1.1.sta0015	04:f0:21:94:98:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.97	0.000000
1.1.sta0016	04:f0:21:94:72:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.13	0.000000
1.1.sta0017	04:f0:21:94:e3:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.25	32.860318
1.1.sta0018	04:f0:21:94:4f:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.01	0.322416
1.1.sta0019	04:f0:21:94:09:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.15	0.271878
1.1.sta0020	04:f0:21:94:28:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.96	0.291645
1.1.sta0021	04:f0:21:94:be:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.07	0.000000
1.1.sta0022	04:f0:21:94:59:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.94	0.494796
1.1.sta0023	04:f0:21:94:ba:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.57	0.653145
1.1.sta0024	04:f0:21:94:d3:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.71	0.757837
1.1.sta0025	04:f0:21:94:d0:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.19	0.241296
1.1.sta0026	04:f0:21:94:64:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.04	0.000000
1.1.sta0027	04:f0:21:94:b0:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.11	0.000000
1.1.sta0028	04:f0:21:94:25:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.48	0.000000
1.1.sta0029	04:f0:21:94:60:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.54	0.000000
1.1.sta0030	04:f0:21:94:ef:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.95	0.000000
1.1.sta0031	04:f0:21:94:56:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.06	0.000000

Information	contact	support@candelatech.com